

Bayesian Learning — Coursework Solutions

Problem 1

Given

Let M = event that Manchester United wins. Let P = event that the pub is packed.

$$P(M) = 0.7$$

$$P(\neg M) = 1 - P(M) = 1 - 0.7 = 0.3$$

$$P(P | M) = 0.9 \quad (\text{probability pub is packed, given United wins})$$

$$P(P | \neg M) = 0.6 \quad (\text{probability pub is packed, given United loses})$$

Bayes' Theorem

$$P(M | P) = [P(P | M) \cdot P(M)] \div P(P)$$

$P(P)$ is unknown, so we find it using the Law of Total Probability. The pub can be packed under two mutually exclusive and exhaustive scenarios — United winning, or United losing:

$$P(P) = P(P | M) \cdot P(M) + P(P | \neg M) \cdot P(\neg M)$$

Calculation

$$\text{Case 1 — United wins: } P(P|M) \cdot P(M) = 0.9 \times 0.7 = 0.63$$

$$\text{Case 2 — United loses: } P(P|\neg M) \cdot P(\neg M) = 0.6 \times 0.3 = 0.18$$

$$\text{Total: } P(P) = 0.63 + 0.18 = 0.81$$

$$\text{Bayes: } P(M | P) = 0.63 \div 0.81 = 0.7778$$

Answer

No, we cannot simply assume United won — a packed pub is also fairly likely (0.6) on a loss, since fans may gather to drown their sorrows rather than celebrate; the observation is evidence, not proof. It does, however, shift belief upward: **$P(M | P) = 0.7778$ ($\approx 77.78\%$)**, up from the 70% prior.

Problem 2

Given

Let F = the nurse forgot the pill, D = Mr. Smith died.

$$P(F) = 0.3, \text{ so } P(\neg F) = 0.7$$

$$P(D \mid \neg F) = 0.1 \text{ (dies if he received the pill)}$$

$$P(D \mid F) = 0.8 \text{ (dies if he did not receive the pill)}$$

Calculation

By the Law of Total Probability:

$$P(D) = P(D \mid F)P(F) + P(D \mid \neg F)P(\neg F) = 0.8 \times 0.3 + 0.1 \times 0.7 = 0.24 + 0.07 = 0.31$$

By Bayes' Theorem:

$$P(F \mid D) = P(D \mid F)P(F) / P(D) = 0.24 / 0.31 = 0.7742$$

Answer

Given that Mr. Smith died, the probability the nurse forgot to give him the pill is **$P(F \mid D) = 0.7742$** ($\approx 77.42\%$). Since missing the pill makes death eight times more likely than receiving it, his death is fairly strong evidence the pill was skipped.

Problem 3

Given

$P(\text{Gold}) = 0.1$, $P(\text{Coal}) = 0.3$, $P(\text{Neither}) = 0.6$ — mutually exclusive and exhaustive (gold and coal cannot both be present).

$$P(\text{Pos} \mid \text{Gold}) = 0.8, P(\text{Pos} \mid \text{Coal}) = 0.4, P(\text{Pos} \mid \text{Neither}) = 0.2$$

Calculation

Law of Total Probability, over the three exhaustive scenarios:

$$P(\text{Pos}) = 0.8 \times 0.1 + 0.4 \times 0.3 + 0.2 \times 0.6 = 0.08 + 0.12 + 0.12 = 0.32$$

Bayes' Theorem:

$$P(\text{Gold} \mid \text{Pos}) = 0.08 / 0.32 = 0.25$$

Answer

Given a positive test, the probability that gold is present on campus is **0.25 (25%)**. The test is more diagnostic of coal or a false positive than of gold here, since gold's low base rate (10%) limits how far even an 80%-accurate positive result can push the posterior.

Problem 4

Given

$P(\text{Meningitis}) = 0.05$, so $P(\neg \text{Meningitis}) = 0.95$

$P(\text{Pos} \mid \text{Meningitis}) = 0.95$ (95% sensitivity)

$P(\text{Neg} \mid \neg \text{Meningitis}) = 0.70 \Rightarrow P(\text{Pos} \mid \neg \text{Meningitis}) = 0.30$ (30% false-positive rate)

4.1 — *Decide from a positive test*

$$P(\text{Pos}) = P(\text{Pos} \mid \text{M})P(\text{M}) + P(\text{Pos} \mid \neg \text{M})P(\neg \text{M}) = 0.95 \times 0.05 + 0.30 \times 0.95 = 0.0475 + 0.285 = 0.3325$$

$$P(\text{M} \mid \text{Pos}) = 0.0475 / 0.3325 = 0.1429$$

$$P(\neg \text{M} \mid \text{Pos}) = 0.285 / 0.3325 = 0.8571$$

Decision: $P(\neg \text{M} \mid \text{Pos}) = 85.71\%$ far exceeds $P(\text{M} \mid \text{Pos}) = 14.29\%$, so the correct decision is that the patient most likely does **not** have Meningitis, despite the positive result. This is because Meningitis is rare (5% prior) while the test's false-positive rate (30%) is high enough that most positives come from healthy patients.

4.2 — *Effect of a second, negative test*

$P(\text{Neg} \mid \text{M}) = 1 - 0.95 = 0.05$, $P(\text{Neg} \mid \neg \text{M}) = 0.70$.

$$P(\text{Neg}) = 0.05 \times 0.1429 + 0.70 \times 0.8571 = 0.00714 + 0.6000 = 0.6071$$

$$P(\text{M} \mid \text{Pos}, \text{Neg}) = 0.00714 / 0.6071 = 0.0118$$

$$P(\neg \text{M} \mid \text{Pos}, \text{Neg}) = 0.9882$$

Effect: belief in Meningitis collapses from 14.29% to **1.18%**, while belief in $\neg$ Meningitis rises to 98.82%. The negative second test strongly reinforces the conclusion from 4.1 that the patient does not have Meningitis, essentially removing any remaining doubt.

Problem 5

Given

$P(\text{Rain tomorrow}) = 0.8$, so $P(\neg\text{Rain}) = 0.2$ (prior).

$$P(\text{Pos} \mid \text{Rain}) = 0.75 \quad P(\text{Pos} \mid \neg\text{Rain}) = 0.15$$

$$\Rightarrow P(\text{Neg} \mid \text{Rain}) = 0.25, \quad P(\text{Neg} \mid \neg\text{Rain}) = 0.85$$

Calculation

$$P(\text{Neg}) = P(\text{Neg} \mid \text{Rain})P(\text{Rain}) + P(\text{Neg} \mid \neg\text{Rain})P(\neg\text{Rain}) = 0.25 \times 0.8 + 0.85 \times 0.2 = 0.20 + 0.17 = 0.37$$

$$P(\text{Rain} \mid \text{Neg}) = 0.20 / 0.37 = 0.5405$$

$$P(\neg\text{Rain} \mid \text{Neg}) = 0.17 / 0.37 = 0.4595$$

Answer

No — Mr. Jones should **not** predict no rain. Even after a negative experiment result, **$P(\text{Rain} \mid \text{Neg}) = 0.5405$ (54.05%)** still exceeds $P(\neg\text{Rain} \mid \text{Neg}) = 45.95\%$. The negative result has weakened his confidence in rain (down from the 80% prior) but not enough to flip the more-likely outcome — rain remains marginally the better prediction.

Problem 6

Given

$P(C) = 0.0001$, so $P(\neg C) = 0.9999$ ($C = \text{Chikungunya}$).

$$P(\text{JointPain} \mid C) = 0.64, \quad P(\text{JointPain} \mid \neg C) = 0.6$$

$$P(\text{Pos} \mid C) = 0.99, \quad P(\text{Pos} \mid \neg C) = 0.04$$

Joint pain and the test result are treated as conditionally independent given disease status.

6.1 — Can the doctor conclude Chikungunya?

$$\text{Numerator}(C) = P(JP|C) \cdot P(\text{Pos}|C) \cdot P(C) = 0.64 \times 0.99 \times 0.0001 = 0.0000634$$

$$\text{Numerator}(\neg C) = P(JP|\neg C) \cdot P(\text{Pos}|\neg C) \cdot P(\neg C) = 0.6 \times 0.04 \times 0.9999 = 0.023998$$

$$P(C | \text{evidence}) = 0.0000634 / (0.0000634 + 0.023998) = \mathbf{0.00263 (0.26\%)}$$

No. Even with a 99%-sensitive positive test, the prior is so low (1 in 10,000) that the posterior barely moves — the doctor cannot conclude Fred has Chikungunya from this alone.

6.2 — Test repeated, second positive

$$P(C | JP, 2 \text{ positives}) = \mathbf{0.0613 (6.13\%)}$$

Belief rises over 20× in relative terms (0.26% → 6.13%) but remains well below 50%, so this alone still does not support a diagnosis.

6.3 — Minimum repeats needed to favour Chikungunya

Each additional positive test multiplies the odds of C by the likelihood ratio $LR = P(\text{Pos}|C)/P(\text{Pos}|\neg C) = 0.99/0.04 = 24.75$. Odds after joint pain alone: 0.0001067.

$$n = 1 \text{ positive: odds} = 0.00264 \Rightarrow P(C) = 0.26\%$$

$$n = 2 \text{ positives: odds} = 0.0653 \Rightarrow P(C) = 6.13\%$$

$$n = 3 \text{ positives: odds} = 1.617 \Rightarrow P(C) = 61.79\%$$

Yes, repeating the test can raise belief above 50% — the minimum is **3 positive results** (i.e. two more tests beyond the first).

Problem 7

Given

10 training records: 5 with Churned = Yes, 5 with Churned = No. Query: Handset = Old, Time > 2.5, Age ≤ 55.

7.1 — Naive Bayes probability of churning

$$\text{Within Yes (n=5): } P(\text{Old}|\text{Yes})=2/5=0.4, P(\text{Time}>2.5|\text{Yes})=2/5=0.4, P(\text{Age}\leq 55|\text{Yes})=2/5=0.4$$

$$\text{Within No (n=5): } P(\text{Old}|\text{No})=0/5=0, P(\text{Time}>2.5|\text{No})=3/5=0.6, P(\text{Age}\leq 55|\text{No})=3/5=0.6$$

$$P(\text{Yes}) \cdot \text{likelihood} = 0.5 \times 0.4 \times 0.4 \times 0.4 = 0.032$$

$$P(\text{No}) \cdot \text{likelihood} = 0.5 \times 0 \times 0.6 \times 0.6 = 0$$

Normalising: **$P(\text{Yes}|\mathbf{X}) = 0.032/(0.032+0) = 1.0$ (100%)**, $P(\text{No}|\mathbf{X}) = 0$. This is a zero-frequency artefact — "Old" simply never appears among the 5 No examples in this small dataset, so the raw estimate is unreliable rather than genuinely certain.

7.2 — *m*-estimate smoothing

m-estimate: $P(\text{attr}=v|\text{class}) = (n_c + m \cdot p) / (n + m)$. Uniform priors: Handset $p=0.5$ (2 values), Time $p=1/3$ (3 values), Age $p=0.5$ (2 values).

With $m = 2$:

$$P(\text{Old}|\text{Yes}) = (2+1)/7 = 0.4286, P(\text{Time}>2.5|\text{Yes}) = (2+0.667)/7 = 0.3810, P(\text{Age}\leq 55|\text{Yes}) = (2+1)/7 = 0.4286$$

$$P(\text{Old}|\text{No}) = (0+1)/7 = 0.1429, P(\text{Time}>2.5|\text{No}) = (3+0.667)/7 = 0.5238, P(\text{Age}\leq 55|\text{No}) = (3+1)/7 = 0.5714$$

$$P(\text{Yes}) \cdot \text{lik} = 0.5 \times 0.4286 \times 0.3810 \times 0.4286 = 0.03498$$

$$P(\text{No}) \cdot \text{lik} = 0.5 \times 0.1429 \times 0.5238 \times 0.5714 = 0.02139$$

$$\mathbf{P(\text{Yes}|\mathbf{X}) = 0.03498/(0.03498+0.02139) = 0.6207 \text{ (62.07\%)}}$$
, $P(\text{No}|\mathbf{X}) = 37.93\%$

Effect of m : larger m pulls every estimate further toward the uniform prior p and away from the raw counts — this fixes the zero-probability problem but dilutes genuine signal from the data. With $m=5$ (checked separately), $P(\text{Yes}|\mathbf{X})$ drops further to ~53.6%, closer to 50/50. As $m \rightarrow 0$ the estimate reverts to the raw MLE (including the zero-frequency issue); as $m \rightarrow \infty$ it approaches the uniform prior regardless of the data — so m controls the trade-off between trusting the (sparse) data and falling back on the prior assumption.

Problem 8

Given

+: (5,6), (5,7), (3,6), (4,5) -: (1,7), (3,4), (2,6), (8,1). X and Y are continuous, assumed independent given class, each Gaussian.

Parameter estimation (sample mean and std, n-1)

Class +: $\mu_x=4.25$, $\sigma_x=0.9574$; $\mu_y=6.00$, $\sigma_y=0.8165$

Class -: $\mu_x=3.50$, $\sigma_x=3.1091$; $\mu_y=4.50$, $\sigma_y=2.6458$

Calculation

$$N(x;\mu,\sigma) = 1/(\sqrt{2\pi}\sigma) \cdot \exp(-(x-\mu)^2/(2\sigma^2))$$

$$N(7|+) = 0.006735, N(4|+) = 0.024326 \Rightarrow \text{joint}(+) = 0.006735 \times 0.024326 = 0.0001638$$

$$N(7|-) = 0.068092, N(4|-) = 0.148117 \Rightarrow \text{joint}(-) = 0.068092 \times 0.148117 = 0.010086$$

$$P(+)\cdot\text{joint} = 0.5 \times 0.0001638 = 0.0000819; P(-)\cdot\text{joint} = 0.5 \times 0.010086 = 0.005043$$

$$P(+ | X=7,Y=4) = 0.0000819 / (0.0000819 + 0.005043) = \mathbf{0.0160 (1.60\%)}$$

Answer

The point (7,4) is classified as negative, with only a 1.60% probability of being positive. It sits far from the '+' cluster's X-mean (4.25) but is well within the wide spread of the '-' cluster ($\sigma_x=3.11$), so the negative class dominates.

Problem 9

Given

Data: 2.8, 4.2, 5.3, 5.5, 2.1. Mixture model: $0.5 \cdot N(\mu_1=2.3, \sigma_1=1.8) + 0.5 \cdot N(\mu_2=6.8, \sigma_2=2.2)$, equal priors.

9.1 — Log-likelihood of the data under this mixture

For each x_i : $p(x_i) = 0.5 \cdot N(x_i; 2.3, 1.8) + 0.5 \cdot N(x_i; 6.8, 2.2)$

$$x=2.8: p=0.12400, \ln p = -2.0876$$

$$x=4.2: p=0.10858, \ln p = -2.2202$$

$$x=5.3: p=0.09950, \ln p = -2.3076$$

$$x=5.5: p=0.09896, \ln p = -2.3130$$

$$x=2.1: p=0.11939, \ln p = -2.1254$$

$$\text{Total log-likelihood} = \sum \ln p(x_i) = \mathbf{-11.054}$$

9.2 — Improved parameter estimates

The given means (2.3 and 6.8) fit the data poorly — all five points lie between 2.1 and 5.5, with nothing near 6.8. A better fit can be obtained by one EM re-estimation (M-step), using each point's responsibility $y_i = P(\text{component 1} \mid x_i)$, already computed in 9.1:

y_1 for $x = 2.8, 4.2, 5.3, 5.5, 2.1$: 0.860, 0.585, 0.278, 0.231, 0.922

New mixing weights: $\pi_1 = \text{mean}(y_1) = 0.575$, $\pi_2 = 0.425$

New means: $\mu_1 = \sum y_{1i} x_i / \sum y_{1i} = 3.318$, $\mu_2 = 4.876$

New std devs: $\sigma_1 = 1.215$, $\sigma_2 = 0.926$

These re-estimated parameters ($\mu_1=3.32$, $\sigma_1=1.21$, $\mu_2=4.88$, $\sigma_2=0.93$, $\pi_1=0.575$) sit much closer to the actual data than the original guess and would yield a strictly higher log-likelihood — this is exactly one iteration of the EM algorithm's M-step, and further iterations would refine it closer to a local optimum.